CHRISTOPHER E. H. LUTSKO

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October 14, 2026

BIOGRAPHICAL

Born: April 26 1994
Pronouns: He/Him
Citizen: USA/UK
Spoken Languages: English, French, Spanish, German.

CAREER

University of Houston 2024 +
Assistant professor

University of Zürich 2023-2024
Postdoctoral researcher

Rutgers University 2020-2023
Hill assistant professor

EDUCATION

University of Bristol 2016-2020
PhD in Mathematics
Advisors: Jens Marklof & Bálint Tóth

University of Texas at Austin 2012-2016
B.S Mathematics
Dean's Scholars Honors Program

PUBLICATIONS

Preprints:

28. *Spacing statistics for a point scatterer on the cubic three-torus*, arXiv:2610.10031 (2026).
27. *Spherical density for real semisimple groups*, arXiv:2610.00298 (2026).
26. *Loss of memory in the periodic Ehrenfest model with small polyhedral scatterers* with J. Marklof, arXiv:2609.04371 (2026).
25. *Poissonian pair correlation for the three-particle Sutherland model*, arXiv:2609.03112 (2026).
24. *Improved sup-norm bounds for locally symmetric spaces*, arXiv:2608.21580 (2026).
23. *Poissonian correlations of* αn^d *mod* 1 with N. Rome and N. Technau, arXiv:2605.06974 (2026).
22. *The Gauss circle problem for Penrose tilings* with A. Haynes, arXiv:2512.21444 (2025).
21. *Average variance bounds for integer points on the sphere*, arXiv:2402.12822 (2024).

Papers:

20. *Exceptional eigenvalue density for thin groups*,
Int. Math. Res. Not. (IMRN), **2026**(18), 208 (2026).

19. *Sign changes along geodesics of modular forms*
with D. Kelmer and A. Kontorovich (2024) [Accepted: **J. Théor. Nombres Bordeaux**].

18. *Counting in lattice orbits*
with A. Kontorovich (2024) [Accepted: **Bull. London Math. Soc.**].

17. *Diffusion of the random Lorentz process in a magnetic field*
with B. Tóth, **J. Math. Phys.**, **66**(11), (2025) (Editor's pick).

16. *Polyhedral bounds on the joint spectrum and temperedness of locally symmetric spaces*
with T. Weich and L. Wolf (2024) [Accepted: **Duke Math. J.**].

15. *An abstract spectral approach to horospherical equidistribution*,
Nonlinearity, **38**(10), 105014, (2025).

14. *Hyperbolic lattice point counting in unbounded rank*
with V. Blomer, **J. Reine Angew. Math.**, **2024**(812), 257-274, (2024).

13. *Norm bounds on Eisenstein series*
with D. Kelmer and A. Kontorovich, **Int. J. Number Theory**, **20** (08), 2083-2098 (2024).

12. *These Numbers Look Random but Aren't, Mathematicians Prove*
Scientific American (2024) (also featured in **Spektrum der Wissenschaft**).

11. *m-Point Correlations of the Fractional Parts of* αn^θ
with N. Technau, **Amer. J. Math.**, **148**(2), 569-597 (2026).

10. *Full poissonian local statistics of slowly growing sequences*
with N. Technau, **Compos. Math., 161**(1), 148-180, (2025).

9. *Effective counting in sphere packings*
with A. Kontorovich, **J. of the Assoc. Math. Res.**, **2**, 15-52 (2024).

8. *Sarnak's spectral gap question*
with D. Kelmer and A. Kontorovich, **J. Anal. Math. 151**, 171-179 (2023) (special volume for P. Sarnak).

7. *Pair correlation of the fractional parts of* αn^θ
with A. Sourmelidis and N. Technau, **J. of the Eur. Math. Soc. (JEMS)**, **27**(10), 4069-4082 (2024).

6. *Long-range correlations of sequences modulo* 1,
J. of Number Theory, **234**, 333-348 (2022).

5. *Farey sequences for thin groups*
Int. Math. Res. Not. (IMRN), **15**, 11642-11689 (2022).

4. *Invariance principle for the random wind-tree process*
with B. Tóth, **Ann. Henri Poincaré**, **22**(10), 3357-3389 (2021).

3. *Invariance principle for the random Lorentz gas – beyond the Boltzmann-Grad limit*
with B. Tóth, **Comm. in Math. Phys.**, **379**, 589-632, (2020).

2. *Directions in orbits of geometrically finite hyperbolic subgroups*,
Math. Proc. of the Cambridge Philos. Soc. 171(2), 277-316 (2021).

1. *A microscopic approach to nonlinear Reaction-Diffusion: the case of morphogen gradient formation*
   with J.P. Boon, J.F. Lutsko, **Phys. Rev. E** 85:021126 (2016).

PhD Thesis:

Statistical Properties of Dynamical Systems: From Statistical Mechanics to Hyperbolic Geometry, 2020 (Bristol University). [Nominated for doctoral dissertation prize]

Conference Proceedings:

1. *Invariance principle for random Lorentz gas in the Boltzmann-Grad Limit*, Oberwolfach Report 10/2019 p. 33-35 (2019)
2. *Invariance principle for random Lorentz gas — Beyond the Boltzmann-Grad Limit*, Oberwolfach Report 42/2019 p. 12-15 (2019)

FUNDING AND AWARDS

Simons Travel Grant (5 years, $35,000)	2025
Nominated for doctoral dissertation prize for outstanding excellence	2020
William's Scholar Honored Graduate	2016

TEACHING

Instructor:

Houston: Math 6322 Functions of Complex Variables	Fall 2026
Houston: Math 2451 Accelerated Calculus	Spring 2026
Houston: Math 3338 Probability	Fall 2025
Houston: Math 2451 Accelerated Calculus	Spring 2025
Houston: Math 2450 Accelerated Calculus	Fall 2024
Rutgers: Math 152 Calculus II (Math/Physics)	Spring 2023
Rutgers: Math 152 Calculus II (Math/Physics)	Spring 2023
Rutgers: Math 356 Theory of Numbers	Fall 2022
Rutgers: Math 437 History of Mathematics	Spring 2022
Rutgers: Math 152 Calculus II (Math/Physics)	Spring 2022
Rutgers: Math 356 Theory of Numbers	Fall 2021
Rutgers: Math 356 Theory of Numbers	Spring 2021
Rutgers: Math 152 Calculus II (Math/Physics)	Spring 2021
Rutgers: Math 152 Calculus II (Math/Physics)	Fall 2020

Teaching Assistant:

Acted as TA for approximately 11 different courses from 2013-2024

POST-DOCS

Kavita Dhanda	2026+

GRADUATE STUDENTS

Anish Ray	2025+
Ziyue Song	2026+

UNDERGRADUATE MENTORSHIP

Arin Morantes (Undergraduate thesis advisor)	Summer 2026+

ORGANIZATIONAL DUTIES

Linfoot number theory seminar (Bristol)	Fall 2019 - Spring 2020
Rutgers number theory seminar	Fall 2021 - Spring 2023
Houston dynamics seminar	Fall 2024 +
Rice-UH Dynamics Day (once per semester)	Spring 2026 +

ACADEMIC INVITATIONS

1 week - Budapest University of Technology (BME)	2017
2 weeks - University of Bonn	2023
1 week - UW Madison	2026

SEMINAR TALKS

University of Bristol	2017
Budapest University of Technology	2017
University of Bristol	2017
University of Warwick	2018
University of Bristol	2018
University of Bristol	2018
Univ. Libre de Bruxelles	2018
University of Bristol	2018
Rutgers University	2019
University of Texas at Austin	2019
University of Houston	2019
University of Exeter	2019
Manchester University	2019
University of Bristol	2019
University of Loughborough	2020
University of Oklahoma	2020
Yeshiva University	2020
Rutgers University	2021
Tata Institute of Fundamental Research	2021
New England Dynamics and Number Theory Seminar (recording: `https://www.youtube.com/watch?v=EODTepggUuU`)	2021
Hong Kong University	2022
University of Illinois Urbana-Champaign	2022
Stony Brook **Colloquium** (recording: `https://www.youtube.com/watch?v=SOHhKdno2jA`)	2022
Rutgers University	2022
New England Dynamics and Number Theory Seminar (recording: `https://nednt.wescreates.wesleyan.edu/fall-2022-schedule/`)	2022
University of Maryland	2022
Philadelphia Area Number Theory Seminar (Bryn Mawr)	2023
Rutgers University	2023
University of Bonn	2023
Paderborn University	2023
University of Zurich	2023
EPFL	2024

University of Bristol 2024
Manchester University (× 2) 2024
University of Houston **Colloquium** 2024
Rice University 2024
One World Probability Seminar 2024
Yale University 2024
University of Houston 2025
IST Austria - University of Graz Joint Seminar 2025
UW Madison 2026
International Seminar on Automorphic Forms 2026
(recording: `https://www.youtube.com/watch?v=PuWv7AB5Tgg`)
Rice University 2026

CONFERENCE TALKS

MINGLE post-graduate event - University of Bristol, UK 2017
Dynamics Days Europe (Billiards Minisymposium) - University of Loughborough, UK 2018
Probability and NonLocal PDEs - University of Swansea, UK 2018
Mini-Workshop: Lorentz Gas Dynamics - M. F. Oberwolfach, Germany 2019
Large Scale Stochastic Dynamics - M. F. Oberwolfach, Germany 2019
MINGLE post-graduate event - University of Bristol, UK 2019
CRC Workshop "Geometry and analysis of locally symmetric spaces" - Paderborn, Germany 2024
Stochastics and Influences - Erdös Center, Budapest, Hungary 2025
Rice-UH Dynamics Day 2026
BIRS Workshop, "Arithmetic, L-functions, and Pseudorandomness" - Banff, Canada 2026

OUTREACH

Volunteer Chemistry Tutor – Garza high school (Austin, Texas) 2015
Help with Bristol maths day 2018
Participant in "Skype a Scientist" 2021
Speaker at Chicago public schools teen mentoring program STEM event 2021
Contributor to EMS magazine problems section 2022
Help with Museum of mathematics' polyplane project 2023
Contributor to Scientific American [link] (this article was also featured in **Spektrum der Wissenschaft**) 2024
Interview on ABC 13 Houston (Houston area television news program) 2025

REFEREEING/REVIEWING/SERVICE

AMS Mathematical Reviews/MathSciNet [12 reviews] • Advances in Mathematics • American Journal of Mathematics • Annales Henri Poincaré • Chaos, Solitons and Fractals • Communications of the AMS • Compositio Mathematica • Ergodic Theory and Dynamical Systems • Indagationes Mathematicae • International Mathematics Research Notices • Journal of the European Mathematical Society • Journal of the London Mathematical Society • Journal of Mathematical Physics • Journal of Number Theory • Journal of Statistical Physics • Mathematical Proceedings of the Cambridge Philosophical Society • Nonlinearity • Random Structures and Algorithms • Research in Number Theory • Stochastic Processes and their Applications